Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/25/2022
PI Name:	Farquharson, Stuart Ph.D.		
Project Title:	Low Gravity Drug Stability Analyzer		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	stu@rta.biz	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	860-635-9800
Organization Name:	Real-Time Analyzers, Inc.		
PI Address 1:	362 Industrial Park Rd, Unit #8		
PI Address 2:			
PI Web Page:			
City:	Middletown	**State:**	CT
Zip Code:	06457-1548	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	05/20/2014	**End Date:**	08/20/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14CA08C		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of this proposed program through Phase III is to build a space-worthy Drug Stability Analyzer that can determine the extent of drug degradation. It will be able to monitor the drug active pharmaceutical ingredient (API) and its degradation product concentrations as a function of time, as well as determine if a drug is suitable for use. This will be accomplished by designing and building a rugged, small, low mass, low power, easy to use analyzer that can identify and quantify API and degradation products with little or no sample handling in 1 minute. Feasibility was successfully demonstrated during Phase I by measuring acetaminophen, azithromycin, epinephrine, lidocaine, and their degradation products in mixtures and during reaction with a 1-4% limit of detection. The APIs were also successfully measured in commercial products. During the Phase II program a prototype Drug Stability Analyzer, suitable for space deployment, will be built and used to measure the degradants of all the International Space Station (ISS) medical kit drugs (>100) with an accuracy goal of 2% and a precision goal of 1% within 1 minute. The Drug Stability Analyzer will be transitioned from a TRL (Technology Readiness Level) 3 to a 7 (ground tested).		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	POTENTIAL NON-NASA COMMERCIAL APPLICATIONS: Approximately 3000 medication errors occur each year due mostly to administering incorrect drugs or doses. Pharmacies are considered the last line of defense and to minimize such errors the Drug Compounding Safety Bill will soon become law. The Drug Stability Analyzer will have great value to pharmacies with its ability to confirm drug identity and dosage in 10 seconds as it is given to the customer. The analyzer will also have great value in its ability to identify counterfeit (fake) drugs. According to the World Health Organization, ~10% of all drugs sold worldwide are counterfeit (largely through the internet), and have resulted in 200,000 deaths per year. Many of the drugs sold in the US are manufactured oversees, and in recent years the standard supply chains have also been compromised.
Task Progress:	New project for FY2014. Reporting not required for this SBIR Phase II project. (Ed. note: added to Task Book on 01/20/2022)
Bibliography Type:	Description: (Last Updated:)